AUTOMATIC BUS SCHEDULING AND ROUTE MANAGEMENT SYSTEM

Submitted By:

Harsh Agnihotri(FH-12515990014)

Kesar Agrawal (FH-2415000805)

Akshat Agrawal (FH-12515990002)

Devansh Chauhan (FH-12515990012)

Jatin Chaudhary (FH-2415000716)

Submitted To:

Mr. Abhishek Singh

(Department Of CEA)

Presentation Agenda

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Current Problems in Delhi Bus System

Manual Scheduling

- Time-consuming process
- Prone to human errors
- Difficult to manage last-minute changes

No Real-Time Tracking

- Endless passenger wait times
- Lack of ETA information
- Imbalanced bus occupancy

Poor Resource Management

- Underutilization of fleet
- Inefficient driver allocation
- Lack of data-driven decision making

System Silos

- Authorities work in disconnected silos
- Depot managers lack digital tools
- Limited communication for drivers
- Passengers lack access to information

OUR SOLUTION

Operations & Scheduling

- Automated scheduling algorithms to replace manual, error-prone processes.
- Digital maintenance tracking with automatic service reminders.
- Data-driven route optimization based on actual passenger demand.

Passenger Experience

- Real-time GPS tracking and accurate arrival predictions.
- Online ticket booking with integrated UPI, credit card, and wallet payments.

Communication & Safety

- Unified platform connecting authorities, depots, and drivers.
- Driver dashboard with an instant SOS button for emergency dispatch.
- One-click delay reporting with automatic passenger notifications.

Administration & Analytics

- Comprehensive analytics dashboard for performance, revenue, and usage patterns.
- Informed decision-making to optimize fleet deployment and service quality.

ONE PLATFORM

Super Admin (Ministry Level)

- City-wide control and analytics
- Policy decisions
- Monitor all transport authorities

Transport Authority (DTC / DIMTS)

- Route management
- Bus allocation
- Operational control

Depot Managers

- Fleet maintenance tracking
- Driver scheduling
- Daily operational oversight

Drivers & Conductors

- Route navigation assistance
- Real-time trip reporting
- Instant emergency dispatch

Passengers

- Live bus tracking
- Bus Reservation
- Route planning

SYSTEM ARCHITECTURE

FRONTEND

- Super Admin Dashboard
- Transport Authority Dashboard
- Depot Manager Portal
- Driver Dashboard
- Passenger Portal

BACKEND

- API Gateway
- Business Logic Layer
- Authentication (JWT)
- Role-Based Access

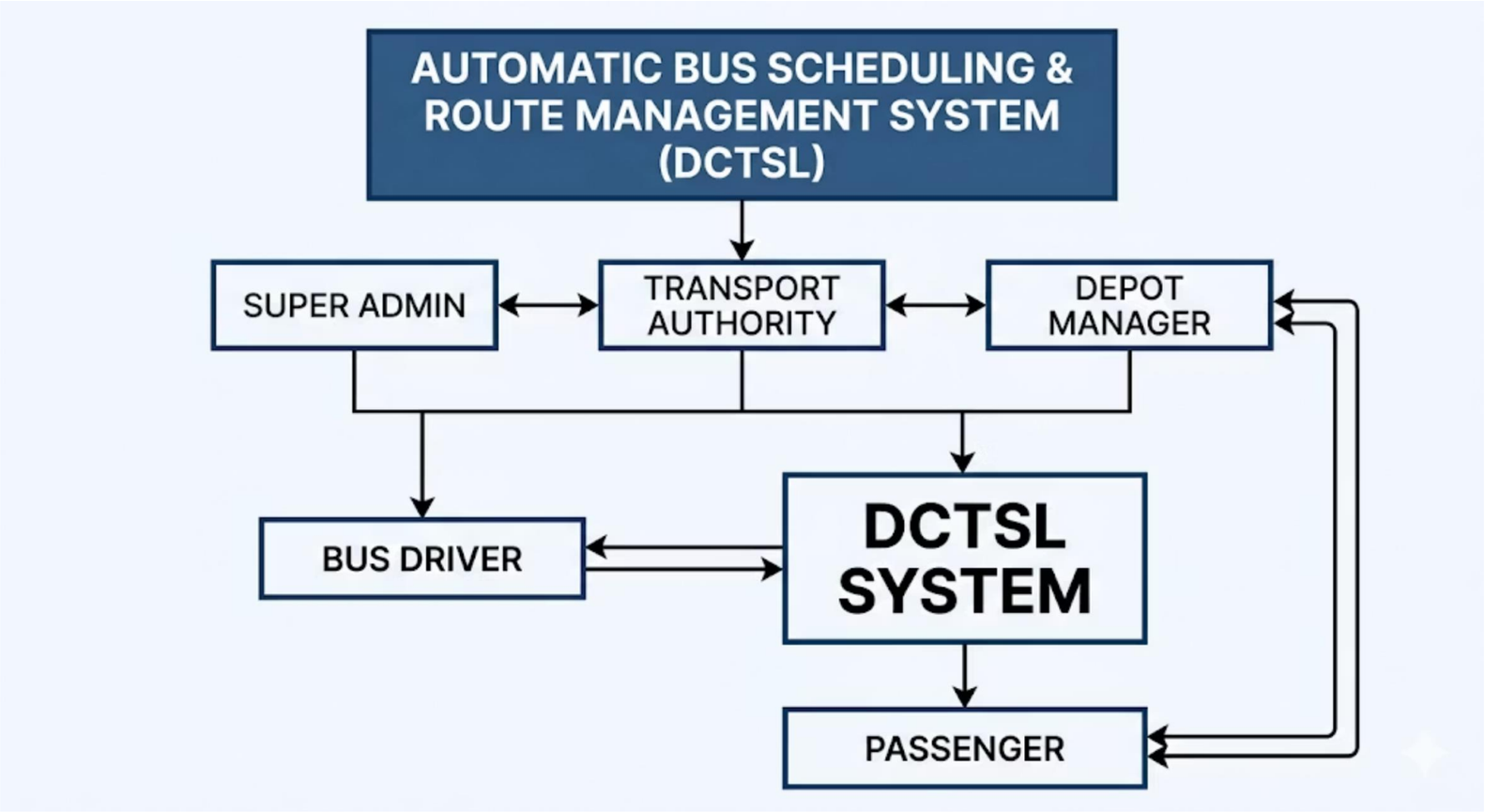
DATABASE

- Users & Roles
- Fleet & Buses
- Routes & Schedules
- GPS Tracking Data
- Bookings

EXTERNAL SERVICES

- **Maps:** OpenStreetMap / Leaflet
- **Payments:** UPI / Card Gateways

DATA FLOW DIAGRAM



Key Features

Multi-Tier Access

Customized dashboards for Super Admins, Transport Authorities, Depot Managers, Drivers, and Passengers with role-specific permissions.

Automated Scheduling

Generates daily timetables automatically, eliminating manual errors and optimizing bus and driver allocation.

Real-Time GPS Tracking

Live bus location monitoring on digital maps with accurate ETAs and route adherence tracking.

Online Booking & Payments

Seamless digital ticketing with support for UPI, cards, and net banking for convenient passenger transactions.

Route Optimization

Data-driven planning to identify high-demand routes and adjust schedules for improved operational efficiency.

How It Works

Our platform streamlines the entire transport management lifecycle through a seamless, five-step integrated workflow.

Step 1: Setup

Super Admins establish authorities, while Depot Managers register essential infrastructure, including buses, routes, and drivers.

Step 2: Scheduling

The system intelligently auto-generates daily schedules by balancing route requirements, vehicle availability, and driver shift constraints.

Step 3: Execution

Drivers initiate trips, activating real-time GPS tracking that provides passengers with live location updates and automated delay updated on app.

Step 4: Booking

Passengers can search for routes, view accurate ETAs, and securely reserve bus using preferred payment methods.

Step 5: Monitoring

Admins gain full operational visibility through real-time dashboards and automated analytics, driving informed decision-making.

Tools & Technologies

Our system leverages a robust stack of modern technologies to ensure performance, scalability, and an optimal user experience.

HTML, CSS, JS

Building dynamic and responsive user interfaces for all dashboards and the passenger portal, ensuring a seamless experience.

Node.js

Powering the backend with efficient, scalable, and high-performance server-side logic and API management.

MongoDB

A flexible NoSQL database to store and manage diverse data, including user profiles, fleet information, routes, and real-time GPS data.

OpenStreetMap / Leaflet

Integrated mapping solutions for real-time bus tracking, route visualization, and optimized navigation features.

UPI / Card Gateways

Secure and convenient online payment processing for ticket bookings, supporting various digital payment methods.

Project Scope

Our comprehensive transportation management system encompasses the following key functionalities:

Operational Efficiency

- **Automated Scheduling:** Daily generation, crew allocation, and manual error reduction.
- **Route Management:** Creation, editing, and optimization based on traffic data.
- **Maintenance:** Service schedules, breakdown reporting, and timely alerts.

System Monitoring

- **Live Tracking:** GPS-based real-time location and accurate ETA calculation.
- **Multi-Role Dashboard:** Dedicated interfaces for Super Admins, Transport Authorities, Depot Managers, Drivers, and Passengers.

User Experience

- **Passenger Features:** Route finding, live tracking, nearby stop info, and feedback systems.
- **Driver Features:** Mobile-friendly interface, one-tap trip management.
- **Booking:** Online Bus reservation with integrated UPI, card, and net banking support.

Team Member Roles

Harsh Agnihotri

DATABASE MANAGER & FRONTEND DEVELOPER

- Design MongoDB schema
- Create collections and relationships
- Optimize queries for fast performance
- Data validation and security

Jatin Chaudhary

FRONTEND DEVELOPER

- Create all UI components
- Build 5 different dashboards
- Responsive design for all devices
- State management (Redux/Context)
- Connect frontend with APIs

Devansh Chauhan

API & GPS TRACKING HANDLER

- Integrate OpenStreetMap/Leaflet
- Implement real-time GPS tracking
- Build location update system
- Calculate ETAs based on traffic
- Handle map visualizations

Kesar Agrawal & Akshat Agrawal

BACKEND DEVELOPERS

- Build REST APIs
- Implement JWT authentication
- Role-based access control
- Business logic implementation
- Server-side validation

Conclusion

This transportation management system represents a significant step toward modernizing urban transit infrastructure. By integrating automated scheduling, real-time monitoring, and user-centric interfaces, the platform addresses critical inefficiencies in current operations.

Operational Impact

Streamlines fleet management and reduces manual intervention through data-driven automation.

Future Potential

Designed for scalability, the system provides a robust foundation for future integrations, including predictive analytics and AI-enhanced route optimization.

Thank You!

Driving the future of urban mobility.

We appreciate your time and interest in our Automatic Bus Scheduling and Route Management System. We are confident this solution can revolutionize urban transportation in Delhi.